Chapter 2

Problem Formulation

2.1 Description of the system dynamics

2.1.1 Kinematics of the motion of the cantilever with end mass

Fig. 2.1 shows a hollow cylinder of mass M, length, L and external diameter D is considered to be the primary system, suspended with the help of a spring. A cantilever beam of length l, density ρ_s, Young's modulus E_s, width, b_S and thickness h_s is fixed inside the hollow cylinder. Another body of mass m is attached at the free end of the cantilever beam. The moment of inertia of this mass with respect to the cantilever is given as J. A representative schematic for describing the geometry of the system is shown in fig 2.2. As shown in the schematic, the primary structure of mass M moves vertically, described by the displacement x_d, and is characterized by stiffness k_1 and a damping constant c_1. A coordinate-system denoted as (x, y), is positioned at the apex of the main structure, where the base of the cantilevered structure is attached. s is the curvilinear coordinate along the beam. The horizontal displacement of a point at a distance s along the cantilever beam is $y(s)$, and the vertical displacement of that point is given by $x(s)$. The slope of that point is given by $\theta(s)$. $\square'$ represents derivative with respect to the spatial coordinate. from the geometry of the problem, $x(s) = \int_0^s \cos(\theta(\eta))\, d\eta \quad and \quad y(s) = \int_0^s \sin(\theta(\eta))\, d\eta$ from this, it can be expressed that:

$$\theta'(s) = \frac{y''(s)}{\sqrt{1-(y'(s))^2}} \approx y''(s)\left(1 + \frac{1}{2}y'(s)^2\right) \tag{2.1}$$

$$x(s) = \int_0^s \sqrt{1-y'(\eta)^2}\, d\eta \approx \int_0^s \left(1 - \frac{(y'(\eta))^2}{2}\right) d\eta \tag{2.2}$$

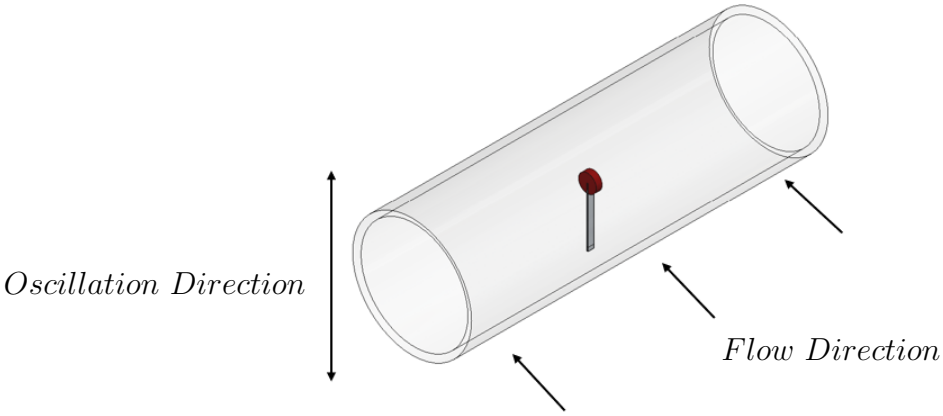

Figure 2.1: The representative CAD model of the bluff body to undergo VIVs

2.1.2 Variational formulation with piezoelectric coupling

The overall kinetic energy (T) is described by adding the energy associated with the translational motion of the primary system, the translational and rotational motion of the cantilever, and the sum of both for the end mass. Mathematically, it is written as:

$$\begin{aligned} T =& \frac{1}{2}M\dot{x}_d^2 + \frac{1}{2}m[\dot{x}_d + \dot{x}(l)]^2 + \frac{1}{2}\int_0^l \rho[\dot{x}_d + \dot{x}(s)]^2 ds \\ &+ \frac{1}{2}m\dot{y}(l)^2 + \frac{1}{2}J\dot{\theta}(l)^2 + \frac{1}{2}\int_0^l \rho\dot{y}(s)^2 ds \end{aligned} \tag{2.3}$$

The total potential energy is the sum of the energy stored in the spring as well as in the of the flexure of the beam.

$$V = \int_0^l \frac{1}{2}EI(\theta'(s))^2 ds + \frac{1}{2}k_2 x_d^2 \tag{2.4}$$

$\dot{\square}$ represents the time derivative. From eq 2.5 and 2.6, the equivalent energy terms can be expressed as

$$\begin{aligned} T =& \frac{1}{2}M\dot{x}_d^2 + \frac{1}{2}m\left(\dot{x}_d - \frac{d}{dt}\int_0^l \frac{y'(\eta)^2}{2}d\eta\right)^2 + \frac{1}{2}\int_0^l \rho\left(\dot{x}_d - \frac{d}{dt}\int_0^s \frac{y'(\eta)^2}{2}d\eta\right)^2 ds \\ &+ \frac{1}{2}m\dot{y}(l)^2 + \frac{1}{2}J\left(y'(l)^2 + \frac{1}{3}y'(l)^4\right) + \frac{1}{2}\int_0^l \rho\dot{y}(s)^2 ds \end{aligned} \tag{2.5}$$

$$V = \int_0^l \frac{1}{2}EI\left(y''(s)^2 + y''(s)^2 y'(s)^2\right) ds + \frac{1}{2}k_2 x_d^2 \tag{2.6}$$

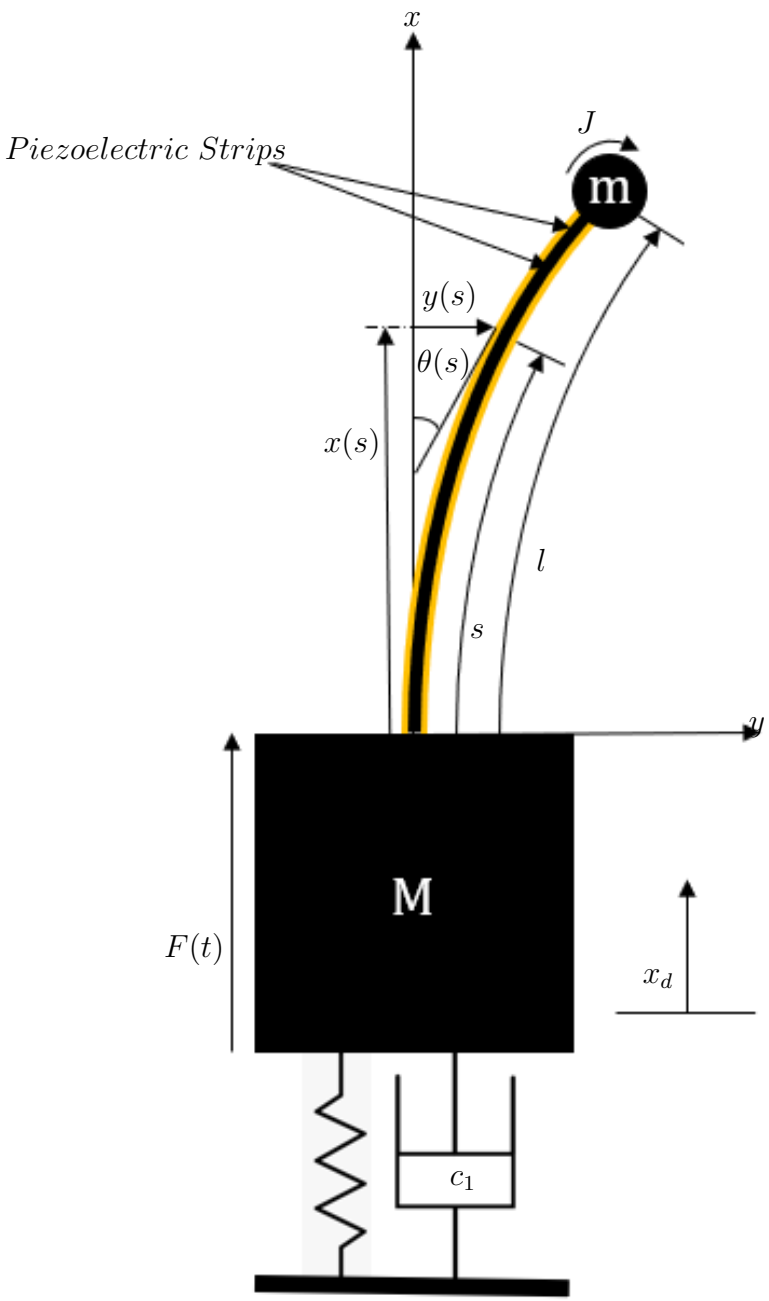

Figure 2.2: A representative schematic of the Autoparametric system

Total non conservative work-done is the sum of the electric work due to the piezoelectric effect, the effect of the dampers, as well as the external force. the total non-conservative work done by the system is given as.

$$W_{nc} = W_{ele} + W_{damp} + W_{ext} \tag{2.7}$$

The direct piezoelectric effect is exploited in order to harvest energy from the system as it is subjected to mechanical deformations. The constitutive equations of the piezoelectric effect describe the mathematical relationship between the electrical and

mechanical responses of the piezoelectric materials. They are described as:

$$\begin{aligned} D_3 &= d_{31}\sigma_1 + \epsilon_{33}^T E_3 \\ S_1 &= s_{11}^E \sigma_1 + d_{13} E_3 \end{aligned} \tag{2.8}$$

where:

$$\begin{aligned} D_3 &= \text{Electric displacement} \\ \sigma_1 &= \text{ Mechanical Stress} \\ E_3 &= \text{Electric Field} \\ S_1 &= \text{Mecanical Strain} \\ d_{31} &= \text{Piezoelectric constant} \\ \epsilon_{33}^T &= \text{Dielectric constant} \\ s_{11}^E &= \text{Elastic compliance} \end{aligned}$$

Considering a bimorph configuration where the cantilever is sandwiched completely between two sheets of a piezoelectric material of width b_p, thickness h_p, density, ρ_p and Youngs modulus E_p. The effective flexural rigidity of the system becomes $EI = \frac{1}{12} b_s h_s^3 E_s + \frac{2}{3} b_p E_p ((h_p + \frac{h_s}{2})^3 - \frac{h_s^3}{8})$ while the mass per unit length is given as $\rho = b_s \rho_s h_s + 2 b_p \rho_p h_p$. If M_{ele} denotes the moment as a consequence of the electric field, given by the expression

$$\begin{aligned} M_{ele} &= b_p \left[\int_{-h_s/2}^{-h_s/2-h_p} e_{31} \left(\frac{V(t)}{h_p} \right) z \, dz + \int_{h_s/2}^{h_s/2+h_p} e_{31} \left(\frac{V(t)}{h_p} \right) z \, dz \right] (\mathbf{H}(s) - \mathbf{H}(s-L)) \\ &= \varphi_p V(t)(H(s) - H(s-L)). \end{aligned} \tag{2.9}$$

Consequently, the virtual work because of the electric field is given as:

$$W_{ele} = \int_0^L -M_{ele} \frac{\partial^2 y(s,t)}{\partial s^2} \, ds \tag{2.10}$$

where $V(t)$ describes the time-dependent output voltage and $\mathbf{H}$ is the heaviside-step function. The term $e_{31} = E_p d_{31}$ denotes the piezoelectric stress coefficient, where E_p denotes Young's modulus of the piezoelectric material, $\varphi_p = -b_2 e_{31}(h_p + h_s)$ gives the piezoelectric coupling term. If the external forcing function is a time dependent $F(t)$ and the damping forces are taken to be $F_{d1} = -c_1 \dot{x}_d$ for the primary, $F_{d2} = -c_2 \dot{y}(s)$ for the secondary cantilever.

$$
\begin{aligned}
W_{damp} &= \oint F_{d1}(t)\, dx_d + \int_0^L \oint F_{d2}(s,t)\, dy\, ds, \\
W_{ext} &= F(t) x_d(t).
\end{aligned}
\tag{2.11}
$$

Applying hamiltons equation by considering $\int_{t_1}^{t_2} (\delta T - \delta V + \delta W_{nc}) dt = 0$, the following equations of motion are obtained

$$
\begin{aligned}
&\rho \ddot{y} + c_2 \dot{y} + EI \left(y'''' + y'^2 y'''' + y''^3 + 4 y' y'' y''' \right) + P y'' \\
&\quad + \left(-\rho \ddot{x}_d + \rho \int_0^s \frac{d^2}{dt^2} \left(\frac{y'^2}{2} \right) d\eta - m \ddot{x}_d \delta(s-l) + m \int_0^l \frac{d^2}{dt^2} \left(\frac{y'^2}{2} \right) d\eta \delta(s-l) \right) y' \\
&\quad + \varphi_p V(t) \left(\frac{\partial \delta(s)}{\partial s} - \frac{\partial \delta(s-l)}{\partial s} \right) = 0.
\end{aligned}
\tag{2.12}
$$

$$
(M + m + \bar{m}) \ddot{x}_d + c_1 \dot{x}_d + k_2 x_d - m \int_0^l \frac{d^2}{dt^2} \left(\frac{y'^2}{2} \right) ds - \int_0^l \rho \int_0^s \frac{d^2}{dt^2} \left(\frac{y'^2}{2} \right) d\eta ds = F(t).
\tag{2.13}
$$

where $\bar{m}$ is equal to $\bar{m} = \int_0^l \rho ds$. P is given as

$$
P = m \ddot{x}_d + \rho \ddot{x}_d (l - s) - m \int_0^l \frac{d^2}{dt^2} \left(\frac{y'^2}{2} \right) d\eta - \int_0^s \rho \int_0^\xi \frac{d^2}{dt^2} \left(\frac{y'^2}{2} \right) d\eta d\xi
\tag{2.14}
$$

The boundary-conditions for this set of equations are:

$$
\begin{aligned}
&y\ (0,t) = 0 \\
&\dot{y}\ (0,t) = 0 \\
&EI \left(y'(l,t)^2 y''(l,t) + y''(l,t) \right) + J \ddot{y}'(l,t) (2(\dot{y}(l,t))^2 + 1) = 0 \\
&EI \left((y'(l,t))^2 y'''(l,t) + y'''(l,t) + y'(l,t)(y''(l,t))^2 + \right) - m \ddot{y}(l,t) = 0
\end{aligned}
\tag{2.15}
$$

The the Gauss law gives the relation between mechanical and electrical variables, given as

$$
\frac{d}{dt} \int_a \mathbf{D} \cdot \mathbf{n}\, dA = \frac{d}{dt} \int_a D_2\, dA = \frac{V}{R}
\tag{2.16}
$$

R denotes the load resistace connected to the circuit. From the constitutive equations and the Euler-Bernoulli beam theory, we get the relation between strain and voltage as:

$$
- e_{31} (h_p + h_s) b_2 \int_0^L \frac{\partial^3 y(s,t)}{\partial t\, \partial s^2}\, ds - \frac{2 \epsilon_{33}^s b_2 L}{h_p} \frac{dV(t)}{dt} = \frac{V(t)}{R}.
\tag{2.17}
$$

2.1.3 Forcing due to the flow stream

To characterize the alternating forces due to the flow stream, a semi-empirical wake oscillator model has been utilized. The main body is taken to be a hollow circular with diameter D and length L cylinder with both ends sealed, subjected to flow of fluid of density ρ_f at a velocity U_∞ along its radial direction. The secondary system of the cantilever beam with endmass is concealed within this cylinder such that it is not exposed to the free stream. The time-dependent forcing function that results in VIV can be defined as:

$$F(t) = \frac{1}{2}\rho_f U_\infty^2 D L C_{x,v} \tag{2.18}$$

The coefficient $C_{x,v}$ is a function of Q_x or the wake variable. The wake oscillator model is applied to solve for this. The equation of the wake oscillator model relates the wake variable to the acceleration of the cylinder as

$$\ddot{Q}_x + \epsilon_x \omega_f (Q_x^2 - 1)\dot{Q}_x + \omega_f^2 Q_x = \frac{A_x}{D}\ddot{x}_d \tag{2.19}$$

Where ω_f is the frequency of vortex-shedding, and A_x, ϵ_x are the empirically determined parameters. $C_{x,v}$ is determined as

$$C_{x,v} = \left(\frac{Q_x C_l^0}{\hat{Q}_x} - \frac{C_d^0 \dot{x}_d}{U_\infty} \sqrt{1 + \left(\frac{\dot{x}_d}{U_\infty}\right)^2} \right) \tag{2.20}$$

where $C_l^0 = 0.3842$ is the lift coefficient, and $C_d^0 = 1.1856$ is the drag coefficient. The empirical parameters are dependent on the reduced velocity $U_r = U_\infty / fD$ such that

$$\epsilon_x = \begin{cases} 0.05, & U_r < 6.5 \\ 0.7, & U_r \geq 6.5 \end{cases} \tag{2.21}$$

$$A_x = \begin{cases} 4, & U_r < 6.5 \\ 12, & U_r \geq 6.5 \end{cases} \tag{2.22}$$

2.1.4 Modal Discretization

To determine the natural frequencies and mode-shapes of the cantilever beam, we discretize the motion of the cantilever beam. The linearized equation of motion is obtained by dropping the non-linear coupling and the damping terms in order to obtain an eigenvalue problem

$$\rho \ddot{y} + EI y'''' = 0 \tag{2.23}$$

then the method of separation of variables is applied to separate out spatial and temporal components of the horizontal displacement as:

$$y(s,t) = \phi(s)q(t) \tag{2.24}$$

Where $\phi(s)$ represents the mode shapes of the structure, which for a 1D beam can be given as:

$$\phi = A_i \sin(\beta s) + B_i \cos(\beta s) + C_i \sinh(\beta s) + D_i \cosh(\beta s); \tag{2.25}$$

The terms A_i, B_i, C_i and D_i, where the subscript i refers to the ith mode, can be determined from the boundary conditions for a cantilever beam. The parameter β is related to the natural frequency ω as $\omega = (\beta^4 EI/\rho)^{\frac{1}{2}}$. Only the first mode of vibration is considered to be relevant to the problem, as the other modes occur at a significantly higher frequency.

In order to determine all the unknowns in the above equation, the following linearized spatial boundary conditions are solved:

$$\begin{aligned} &\phi(0) = 0 \\ &\phi'(0) = 0 \\ &EI\phi''(l) - \omega^2 J\phi'(l) = 0 \\ &EI\phi''(l) + \omega^2 m\phi(l) = 0 \end{aligned} \tag{2.26}$$

The following orthogonality conditions also need to be satisfied in order to find the normalized mode shapes.

$$\begin{aligned} &\int_0^l m\phi_b(l)\phi_r(l) + \rho\phi_b(s)\phi_r(s)\, ds + J\phi_b'(l)\phi_r'(l) = \delta_{br} \\ &\int_0^l EI\phi_r''(s)\phi_b''(s)\, ds = \delta_{rb}\omega_r^2 \end{aligned} \tag{2.27}$$

δ represents the Kronecker delta.

2.1.5 Governing equations of the discretized system coupled with the wake oscillator model

Following the separation of variable method and substituting $y(s,t) = \phi(s)q(t)$, denoting $\frac{2\epsilon_{33}^s b_2 L}{h_p}$ as capacitance C_p and introducing $\theta_p = \phi'(L)\varphi_p$, we get the following governing equations

$$\ddot{q}_d + \eta_1\dot{q}_d + \omega_1^2 q_d + \eta_2 q_d^3 + \eta_3 q_d(\dot{q}_d^2 + q_d\ddot{q}_d) - \eta_4\ddot{x}_d q_d + \theta_p V(t) = 0. \tag{2.28}$$

$$(M+m+\bar{m})\ddot{x}_d + c_2\dot{x}_d + k_2 x_d - \eta_4\left(\dot{q}_d^2 + q_d\ddot{q}_d\right) = \frac{1}{2}\rho_f U_\infty^2 DL\left(\frac{Q_x C_l^0}{\hat{Q}_x} - \frac{C_d\dot{x}_d}{U_\infty}\sqrt{1+\left(\frac{\dot{x}_d}{U_\infty}\right)^2}\right). \tag{2.29}$$

$$\frac{V(t)}{R} + C_p\dot{V}(t) - \theta_p\dot{q}_d(t) = 0. \tag{2.30}$$

$$\ddot{Q}_x + \epsilon_x\omega_f(Q_x^2-1)\dot{Q}_x + \omega_f^2 Q_x = \frac{A_x}{D}\ddot{x}_d. \tag{2.31}$$

where in these equations;

$$\begin{aligned}
\eta_1 &= \int_0^l c_2\phi^2 ds,\\
\eta_2 &= \int_0^l EI\left(\phi''''\phi'^2\phi + \phi\phi'^3 + 4\phi\phi'\phi''\phi'''\right)ds\\
\eta_3 &= m\left(\int_0^l \phi'^2 ds\right)^2 + \int_0^l \rho\left(\int_0^s \phi'(\eta)^2 d\eta\right)^2 ds\\
\eta_4 &= m\int_0^l \phi'^2 ds + \int_0^l \rho\int_0^s \phi'(\eta)^2 d\eta ds
\end{aligned} \tag{2.32}$$

If we choose X_0 as the small variable to non-dimensionalize the variables, the equations can be represented as:

$$\ddot{x} + 2\epsilon\mu_2\dot{x} + \omega_1^2 x - \epsilon\left(q\ddot{q} + \dot{q}^2\right) = \frac{1}{2(M+m+\bar{m})}\rho_f U_\infty^2 DL\left(\frac{QC_l^0}{\hat{Q}} - \frac{C_d\dot{x}}{U_\infty}\sqrt{1+\left(\frac{\dot{x}_d X_0}{U_\infty}\right)^2}\right) \tag{2.33}$$

$$\ddot{q} + 2\epsilon\mu_1\dot{q} + \omega_{beam}^2 q + \delta_1\epsilon^2 q^3 + R\epsilon^2 q\left(q\ddot{q} + \dot{q}^2\right) - \epsilon\ddot{x}q + \theta_p V_n(t) = 0 \tag{2.34}$$

$$\frac{V_n(t)}{R} + C_p\dot{V}_n(t) - \theta_p\dot{q}(t) = 0 \tag{2.35}$$

$$\ddot{Q} + \epsilon_x\omega_f((QX_0)^2-1)\dot{Q} + \omega_f^2 Q = \frac{A_x}{D}\ddot{x} \tag{2.36}$$

where:

$$q = \frac{q_d}{\sqrt{M+m+\bar{m}}X_0}, \quad x = \frac{x_d}{X_0}$$
$$V_n = \frac{V}{\sqrt{m+M+\bar{m}}X_0}, \quad Q = \frac{Q_x}{X_0}$$
$$\omega_1 = \sqrt{\frac{k_1}{m+M+\bar{m}}}$$
$$\omega_{beam} = \text{natural frequency of the secondary system}$$
$$\mu_1 = \frac{\eta_1}{2\epsilon}$$
$$\mu_2 = \frac{c_2}{2\epsilon(M+m+\bar{m})}$$
$$R = \frac{m+M+\bar{m}}{m}$$
$$\epsilon = X_0\eta_4$$
$$\delta_1 = \frac{(M+m+\bar{m})\eta_2}{\eta_4^2}$$

2.1.6 Formulation with two secondary systems

Consider a similar second cantilever beam with an end mass added to the primary system in a similar way as the first one (Refer fig 2.3).

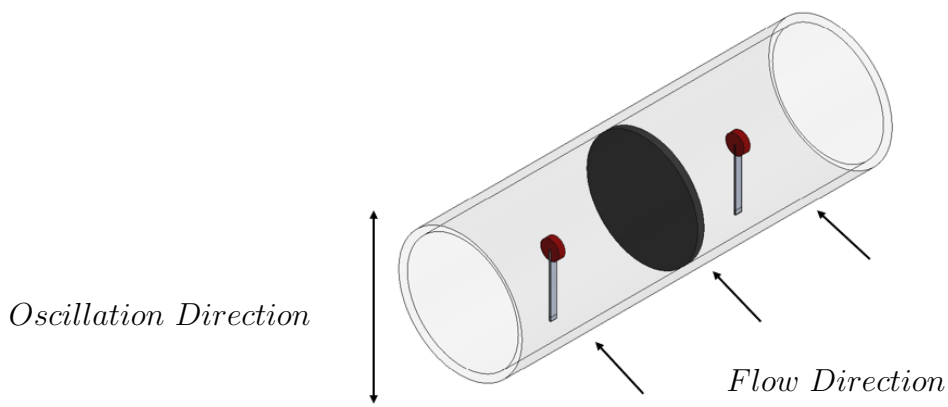

Figure 2.3: The representative CAD model of the bluff body with two secondary systems

Now, the first beam has length l_1 and end mass, m_1 and the second beam has length l_2 and end mass m_2. Both vibrate in a transverse fashion in different parallel planes such that there is no interaction between the beams when they oscillate. Deriving the

governing equations in the same way as described previously, we get:

$$\ddot{x} + 2\epsilon_{21}\mu_1\dot{x} + \omega_1^2 x - \epsilon_{21}\left(q_1\ddot{q}_1 + \dot{q}_1{}^2\right) - \epsilon_{22}\left(q_2\ddot{q}_2 + \dot{q}_2{}^2\right) = \frac{1}{2(M + m_1 + m_2 + \bar{m}_1 + \bar{m}_2)}\rho_f U_\infty^2 DL\left(\frac{QC_l^0}{\hat{Q}} - \frac{C_d\dot{x}}{U_\infty}\right)\sqrt{1 + \left(\frac{\dot{x}_d X_0}{U_\infty}\right)^2}. \tag{2.37}$$

$$\ddot{q}_1 + 2\epsilon_{21}\mu_{21}\dot{q}_1 + \omega_{beam1}^2 q_1 + \delta_{21}\epsilon_{21}^2 q_1^3 + R_1\epsilon_{21}^2 q_1\left(q_1\ddot{q}_1 + \dot{q}_1{}^2\right) - \epsilon_{21}\ddot{x}q_1 = 0. \tag{2.38}$$

$$\ddot{q}_2 + 2\epsilon_{22}\mu_{22}\dot{q}_2 + \omega_{beam2}^2 q_2 + \delta_{22}\epsilon_{22}^2 q_2^3 + R_2\epsilon_{22}^2 q_2\left(q_2\ddot{q}_2 + \dot{q}_2{}^2\right) - \epsilon_{22}\ddot{x}q_2 = 0. \tag{2.39}$$

$$\frac{V_{n1}(t)}{R} + C_p\dot{V}_{n1}(t) - \theta_{p1}\dot{q}_1(t) = 0. \tag{2.40}$$

$$\frac{V_{n2}(t)}{R} + C_p\dot{V}_{n2}(t) - \theta_{p2}\dot{q}_2(t) = 0. \tag{2.41}$$

$$\ddot{Q} + \epsilon_x\omega_f((QX_0)^2 - 1)\dot{Q} + \omega_f^2 Q = \frac{A_x}{D}\ddot{x}. \tag{2.42}$$

where V_{n1} is the normalized voltage output of the first system and V_{n2} is the voltage

output of the second system the rest of the parameters can be defined similarly as:

$$\omega_1 = \sqrt{\frac{k_2}{M + m_1 + m_2 + \bar{m_1} + \bar{m_2}}}$$

ω_{beam1} = fundamental frequency of the first cantilever with end mass

ω_{beam2} = fundamental frequency of the second cantilever with end mass

$$\mu_1 = \frac{c_2}{2\epsilon(M_{tot})}$$

$$\mu_{21} = \frac{\eta_{211}}{2\epsilon_1}$$

$$\mu_{22} = \frac{\eta_{221}}{2\epsilon_2}$$

$$R_1 = \frac{M_{tot}}{m_1}$$

$$R_1 = \frac{M_{tot}}{m_2}$$

$$\epsilon_{21} = X_0\eta_{214}$$

$$\epsilon_{22} = X_0\eta_{224}$$

$$\delta_{21} = \frac{(M_{tot})\eta_{212}}{\eta_{214}^2}$$

$$\delta_{22} = \frac{(M_{tot})\eta_{222}}{\eta_{224}^2}$$

$\eta_{211}, \eta_{212}, \eta_{213}, \eta_{214}, \eta_{221}, \eta_{222}, \eta_{223}$ and η_{224} can be calculated in a similar way as described earlier with ϕ_1 taken as the mode shape equation for the first beam and ϕ_2 as the mode shape equation for the second beam.

An analysis of the response of the system with respect to change in the forcing frequency is carried out for various cases.The frequency of the beam can be varied by changing the beam parameters. For the purpose of the same, the following parameters are described as:

Parameter	**Definition**
α	ω_1/ω_{beam1}
β	ω_1/ω_{beam2}
r	Ω/ω_1

Table 2.1: Description of variable parameters

The variation in these 3 parameters lets us analyze how the system behaves with respect to the autoparametric interactions between the secondary system(s) and the primary system.